Identifying Sales Drivers and Growth Opportunities in Retail Store Performance

Executive Summary

This analysis evaluates weekly sales performance across 45 stores to identify key drivers, growth opportunities, and sources of variability. Results indicate that seasonal factors, particularly holiday periods, are the primary driver of sales, while store-level performance differences and growth trends present the most actionable opportunities for optimization.

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Introduction

Overview

Sales performance at the store level is influenced by a wide range of factors, including seasonal patterns, operational dynamics, and broader economic conditions. Identifying which of these factors has the greatest impact enables more effective decision making, allowing organizations to capitalize on favorable trends and mitigate potential declines. This analysis examines multiple potential drivers of sales performance across a network of stores, with the objective of quantifying the extent to which these factors influence sales outcomes and identifying actionable patterns that can inform strategic decisions.

The analysis begins with an evaluation of overall sales trends over time, both at the aggregate level and across individual stores, to identify seasonality and relative performance differences. Building on this foundation, store-level growth rates are calculated to assess performance trajectories and categorize stores based on their growth characteristics. Next, a forecasting model is developed to evaluate sales predictability and consistency across stores, with particular attention given to identifying locations with stable or highly variable performance. The analysis then examines the impact of holiday periods to determine the extent to which seasonal events influence sales patterns. Finally, the relationship between sales and selected economic factors, including Consumer Price Index (CPI), unemployment rate, fuel price, and temperature, is evaluated to assess whether external macroeconomic conditions contribute meaningfully to sales performance.

Data

The dataset used in this analysis contains weekly sales data for 45 stores spanning the period from February 5th, 2010, to October 26th, 2012. In addition to sales figures, the dataset includes weekly observations of Consumer Price Index (CPI), unemployment rate, fuel price, and temperature corresponding to each store's region. The dataset is intentionally scoped to focus on high-level drivers of sales performance. It does not include product-level detail, category breakdowns, or demographic and geographic characteristics of store locations. These factors are considered outside the scope of the current analysis but represent potential areas for future investigation.

Sales Trends

Analysis

We evaluate both aggregate and store-level average sales to identify overarching patterns and potential seasonality in performance. This includes examining trends over time, as well as identifying weeks and stores corresponding to the highest and lowest sales outputs. At the aggregate level, we analyze total weekly sales across all stores to detect seasonal patterns. At the store level, we compute average sales per store over the full observation period to assess relative performance across locations.

Results

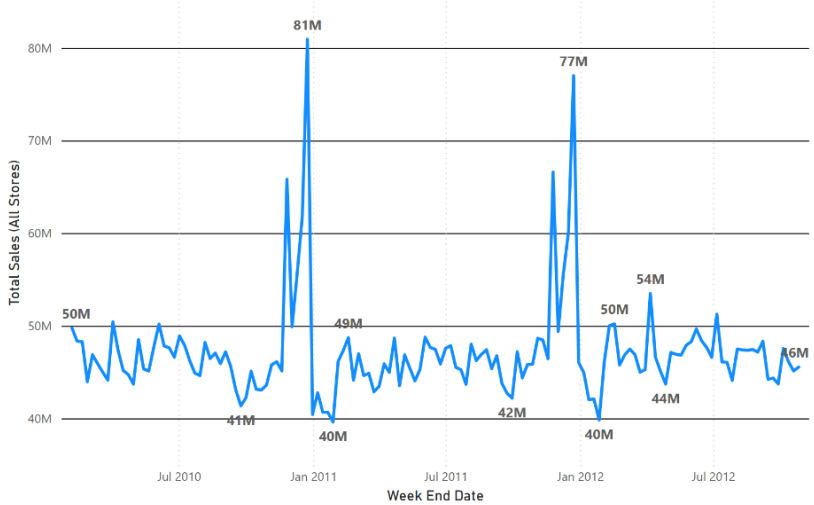

Figure 1: Overall Sales Over Time

The most prominent pattern is the presence of recurring spikes in sales toward the end of each calendar year. Excluding these peak periods, average weekly sales are $46,380,393.68. During the peak periods, average sales increase to $72,586,066.35, representing a 56.5% increase. **Figure 1** illustrates total weekly sales across all stores over the analysis period. The timing of these spikes suggests a strong seasonal component, potentially driven by holiday-related consumer behavior. Outside of these seasonal surges, weekly sales remain relatively stable. The standard deviation of weekly sales is $3,148,375.40, which is less than 7% of the mean, indicating low variability during non-peak periods.

The highest sales weeks occur consistently near year-end, with the highest performing week in the analysis period ending on 12/24/2010 and having total sales of $80,931,415.60. The remainder of the top five highest performing sales weeks are those ending on 12/23/2011 ($76,998,241.31), 11/25/2011 ($66,593,605.26), 11/26/10 ($65,821,003.24), and 12/17/2010

($61,820,799.86). In contrast, the lowest sales weeks tend to occur early in the calendar year, specifically during the weeks ending on 01/28/2011 ($39,599,852.99), 01/27/2012 ($39,834,974.67), 12/31/2010 ($40,432,519.00), 01/21/2011 ($40,654,648.03), and 01/14/2011 ($40,673,678.04). The median weekly sales value occurs during the week ending on 07/16/2010, with total sales of $46,243,899.58, which is closely aligned with the non-peak average.

The highest performing individual store is number 20, with an average sales figure of $2,107,676.87 in our range of dates. Other top performing stores include stores numbered 04 ($2,094,712.96), 14 ($2,020,978.40), 13 ($2,003,620.31), and 02 ($1,925,751.34). At the lower end of the distribution, the five lowest performing stores are stores numbered 33 ($259,861.69), 44 ($302,748.87), 05 ($318,011.81), 36 ($373,511.99), and 38 ($385,731.65). The median store is number 34, with an average sales figure of $966,782.00. **Figure 2** presents the average sales by store over the full time period, organized by store number. This view highlights substantial variation in performance across locations.

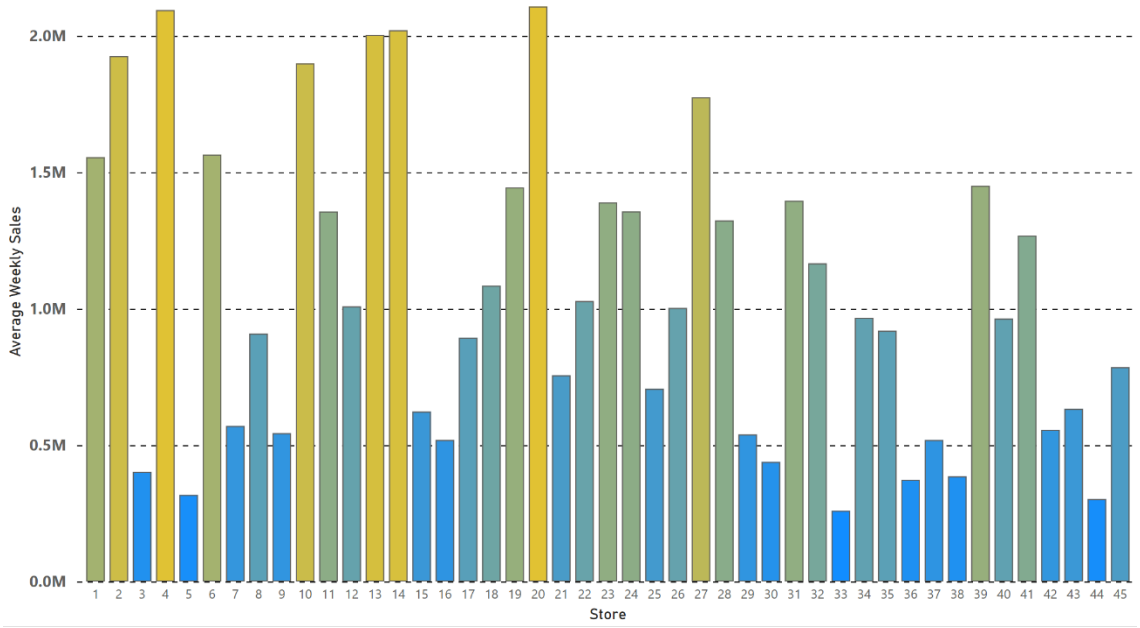

Figure 2: Average Sales by Store

Overall, the distribution of average sales by store indicates a significant performance gap between highest and lowest performing locations, suggesting that store-level factors may play a meaningful role in driving sales outcomes.

Growth Categorization

Analysis

To assess store-level performance trends over time, we estimate a linear trend for each store's weekly sales and use the slope of this trendline as a proxy for growth rate. A positive slope indicates that a store's sales are increasing over time, while a negative slope indicates declining performance.

To systematically classify stores, we benchmark each store's growth rate against the overall distribution of growth rates across all 45 stores. The mean growth rate is approximately 95.00, indicating that, on average, stores experience a weekly increase of $95 in sales. We define classification thresholds using the standard deviation of growth rates. A growth store is defined as a store with a growth rate greater than 0.5 standard deviations above the mean, while a declining store is defined as one with a growth rate more than 0.5 standard deviations below the mean. This approach allows us to identify stores with meaningfully above- or below-average performance trends while accounting for overall variability in the data.

Results

Table 1 Highest Growth Stores

Store	Growth Rate
39	2,371.14
4	2,305.07
41	1,915.71
13	1,182.74
17	1,076.26

Using the defined thresholds, we identify 9 stores that mee the criteria for growth. Among these, the stores with the strongest upward trends are those numbered 39, 04, 41, 13, and 17, with their corresponding growth rates presented in **Table 1**. The remaining growth stores are those numbered 07, 38, 01, and 32.

Table 2 Highest Decline Stores

Store	Rate of Decline
14	-3,275.05
35	-2,271.20
36	-1,381.60
27	-1,222.67
10	-896.72

Using the same thresholding approach, we identify 11 stores that meet the criteria for decline. The stores with the most pronounced negative trends are those numbered 14, 35, 36, 27, and 10, with their rates of decline shown in Error! Reference source not found.. The median growth rate is observed in store number 11, with a value of approximately 208.53, indicating that the distribution of growth rates is slightly right-skewed relative to the

mean. A complete list of all stores and their corresponding growth rates is provided in **Table 3** in the appendix.

Forecasting and Sales Predictability

Analysis

To evaluate sales predictability across stores, we develop a two year forecast using a random forest model. Model performance is assessed at the store level using R^2 as a measure of explanatory power and volatility as a measure of variability in sales. The random forest model was selected based on its superior performance relative to alternative models considered, as it produced the lowest error among all candidates. A comparison of model performance metrics is provided in **Table 4** in the appendix.

In addition to evaluating model performance, we classify stores based on their R^2 scores and volatility levels. These classifications are used to identify stores with consistently predictable sales patterns as well as those with higher uncertainty. Finally, we generate and visualize forecasted sales for the subset of stores previously identified as having the highest growth rates.

Results

The model demonstrates strong predictive performance for several stores. The highest R^2 score is observed at store number 10, with a value of 0.95, indicating that the model performs exceptionally well at predicting that store's sales. Other high-performing stores include those numbered 22 (0.94), 36 (0.93), 38 (0.86), and 07 (0.85).

At the lower end, several stores exhibit poor model fit. Store number 11 has an R^2 score of -3.42, indicating that the model performs substantially worse at predicting that store's sales than a simple mean-based prediction. Additional low-performing stores include those numbered 27 (-0.66), 43 (-0.52), 12 (-0.44), and 45 (0.14).

To classify stores, threshold values are defined using the 25th and 75th percentiles for both metrics. Stores that exceed the upper quartile for R^2 while remaining below the lower quartile for volatility are classified as highly predictable. Conversely, stores that fall below the lower quartile for R^2 and exceed the upper quartile for volatility are classified as least predictable. **Figure 3** presents a joint view of R^2 scores and volatility across stores, with volatility plotted on the x-axis and R^2 on the y-axis.

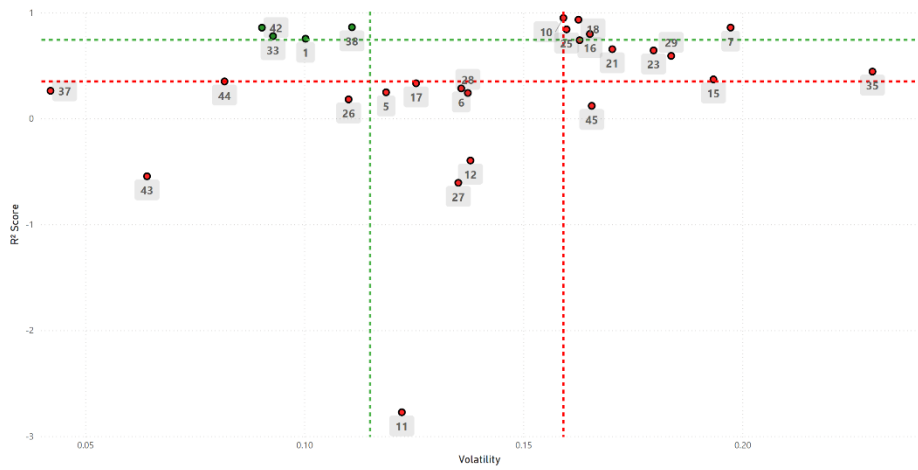

Figure 3 R^2 Score and Volatility by Store

Using this framework, stores numbered 01, 13, and 42 meet the criteria for high predictability. In contrast, store number 45 is the only store that meets both criteria for low predictability. Several stores meet only one of the benchmark conditions, indicating partial alignment with either strong model fit or low variability but not both. All remaining stores fall within the interquartile range for both metrics and are therefore not classified as extreme in either direction.

displays the forecasted sales trajectories for the previously identified high-growth stores. A black dashed line distinguishes observed data from the forecasts generated by our model. Across the stores, the model projects a general downward trend in sales over the forecast horizon. The stores exhibiting the smallest projected declines are those numbered 30, 29, 38, 36, and 37. In contrast, the steepest projected declines are observes in stores numbered 06, 02, 11, 12, and 01.

When applying the decline threshold defined in the Growth Categorization section, only two stores remain above this threshold and are therefore considered stable within the forecast period. All other stores meet the criteria for decline based on their projected trends.

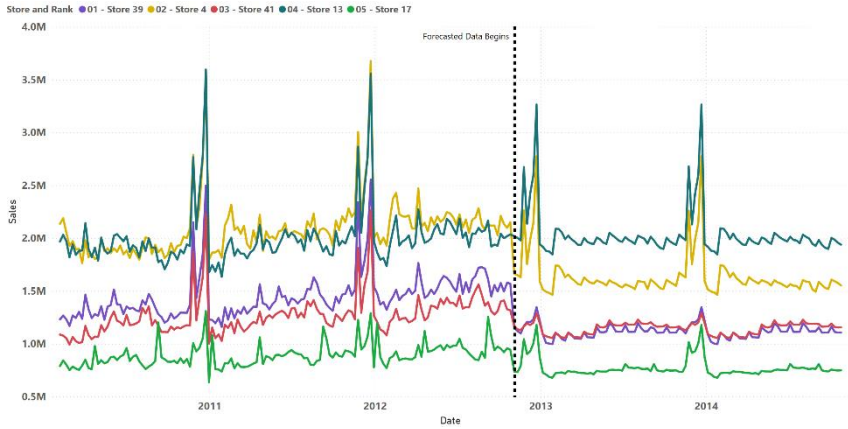

Figure 4 Sales of Highest Growth Stores Over Time (with Forecasted Data)

Holiday Impact on Sales

Analysis

To evaluate the impact of seasonality on sales performance, we compare average weekly sales during holiday periods to those observed during non-holiday periods. For this analysis, a holiday week is defined as any week that either contains a recognized holiday or immediately precedes a week containing a major holiday. Within the scope of this dataset, the only data classified as a major holiday is December 25th of each year. This definition allows us to capture both direct holiday effects and anticipatory consumer behavior leading into major retail events.

Results

We observe a clear and consistent increase in sales during holiday periods. Average weekly sales during non-holiday weeks are $46,366,334.10, while holiday weeks produce average sales of $55,269,100.80. This represents an increase of 19.2%, indicating a substantial seasonal uplift.

At the store level, the magnitude of the holiday effect varies considerably. The largest increase is observed with store number 07, which experiences a sales uplift of approximately 35.25% during holiday weeks. Other stores with similarly strong holiday performance include stores numbered 15 (34.56%), 35 (34.13%), 29 (32.51%), and 21 (28.61%). In contrast, a small subset of stores exhibits minimal or even negative holiday impact. Store number 38 experiences the largest decline in sales during holiday periods, with an observed decrease of approximately 2.77%. Additional stores with negative or low holiday uplift are stores numbered 36 (-0.54%), 33 (-0.36%), 42 (0.22%), and 37 (0.43%). **Figure 5** presents the percentage change in sales for each store during holiday weeks relative to non-holiday periods.

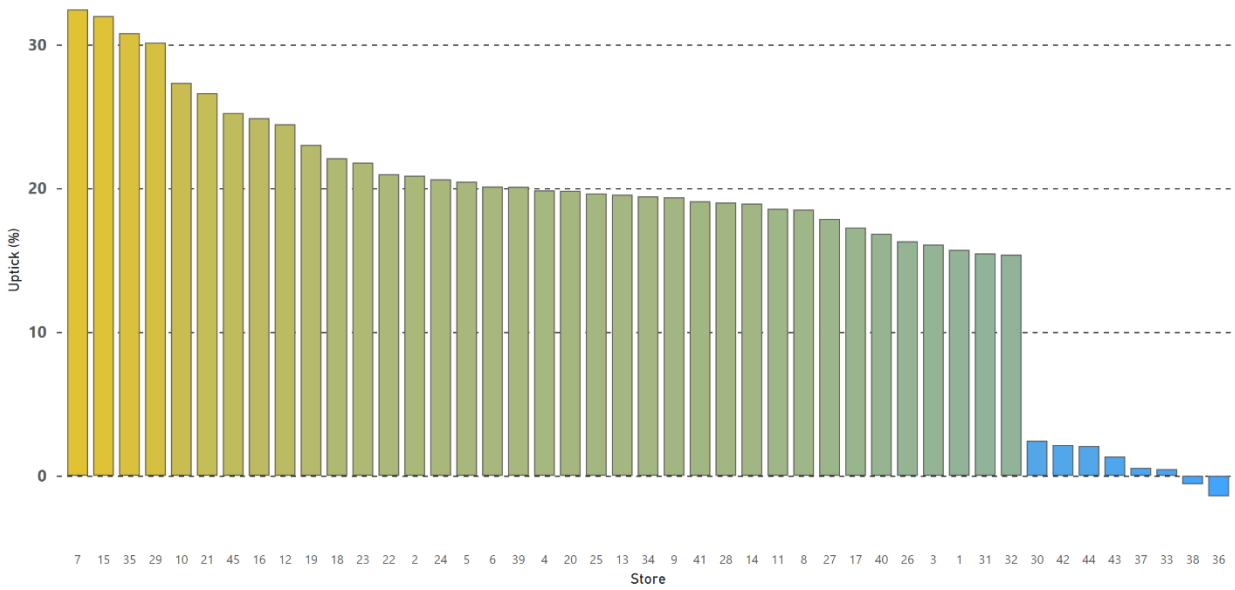

Figure 5 Holiday Uptick by Store (Percentage)

Overall, while the majority of stores benefit from increased consumer activity during holiday periods, the variability in uplift across stores suggests that local factors or operational differences may influence the extent to which individual locations capitalize on seasonal demand.

Economic Factors

Analysis

To evaluate the relationship between macroeconomic conditions and sales performance, we analyze four variables: Consumer Price Index (CPI), unemployment rate, fuel price, and

temperature. For each factor, we construct scatter plots comparing that factor's value to weekly sales at the store level. Because these variables vary by store location, the analysis is conducted at the store level rather than in aggregate. For each scatter plot, we estimate a trendline to identify any relationship between the factor and sales.

In addition to visual analysis, we compute correlation coefficients to quantify the strength and direction of the relationship between each factor and sales. These values provide a standardized measure for comparing the relative influence of each variable. We also perform a lag analysis to determine whether changes in these factors have a delayed effect on sales performance. This involves measuring the impact of a change in each variable on sales in subsequent weeks.

Results

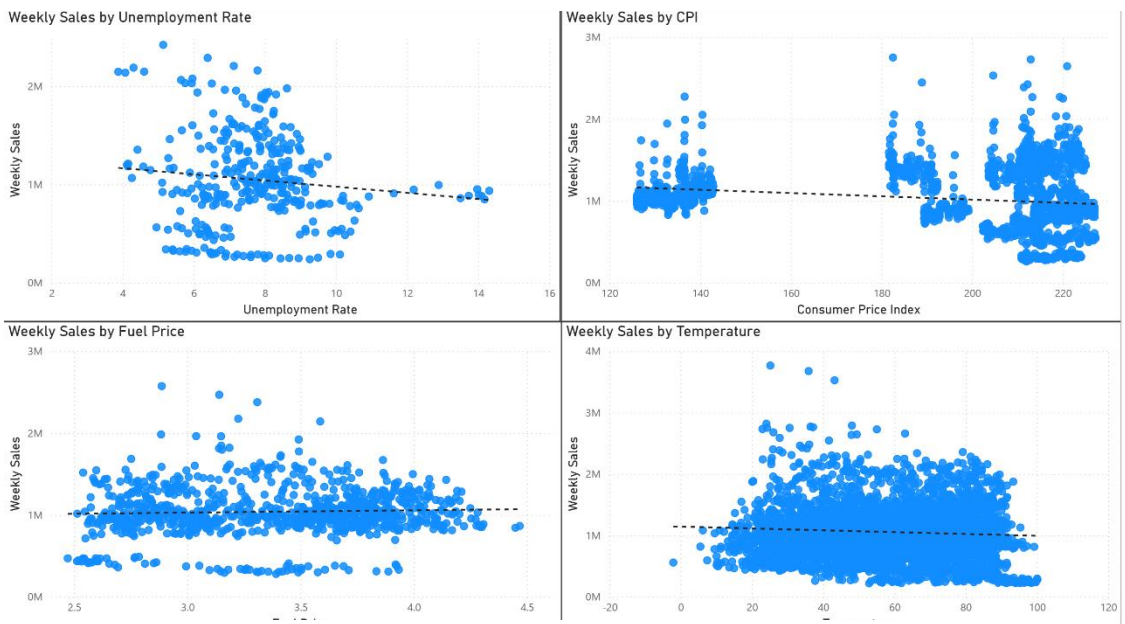

Figure 6 Scatter Plots of Economic Factors with Sales

Visual inspection of the scatter plots in **Figure 6** indicates no strong or consistent relationship between any of the examined factors and sales performance. While slight patterns are observable in some cases, none suggest a meaningful or actionable trend. This conclusion is supported by the calculated correlation coefficients show in **Figure 7**. Among the variables analyzed, unemployment rate exhibits the strongest relationship with sales, with a correlation value of -0.11. For context, a correlation value of 1 represents a perfect positive relationship, a value of -1 represents a perfect inverse relationship, and a value of 0 indicates no linear

relationship. The remaining variables show similarly weak associations with sales. CPI has a correlation value of -0.07, temperature has a correlation value of -0.06, and fuel price has a correlation value of 0.01. These values further reinforce the absence of any meaningful linear relationship between the selected economic factors and sales performance.

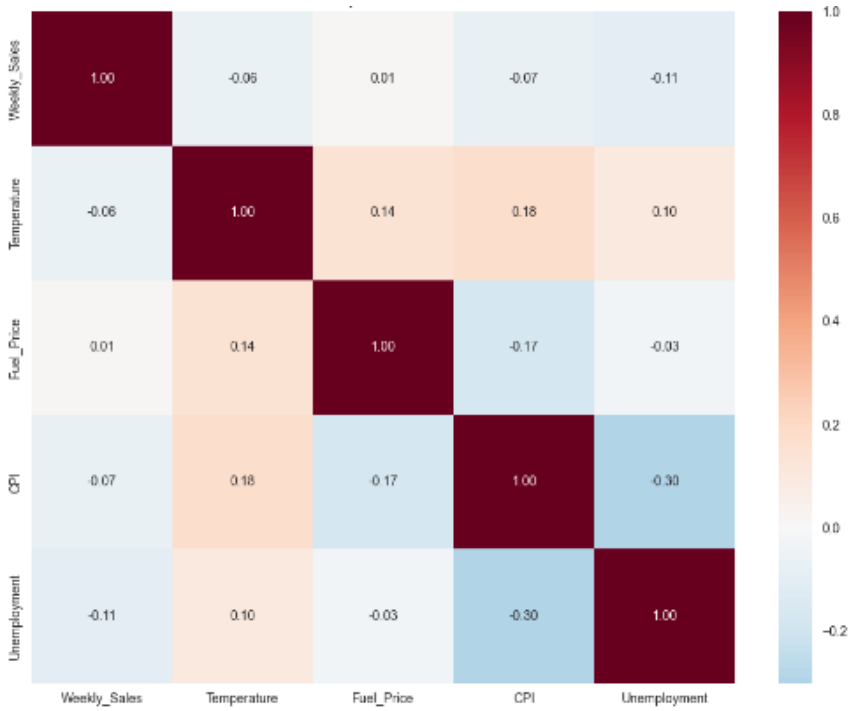

Figure 8 Correlation Heatmap of Economic Factors and Sales

The lag analysis also indicates minimal delayed effects. Across all factors and stores, changes in economic conditions result in only negligible subsequent changes in sales. The most pronounced lagged effect is observed with store number 35, where a 1% increase in CPI is associated with an approximate 0.64% decrease in sales two weeks later. Even in this cast, the magnitude of the effects remains small. **Figure 8** presents a detailed view of the CPI lag analysis,

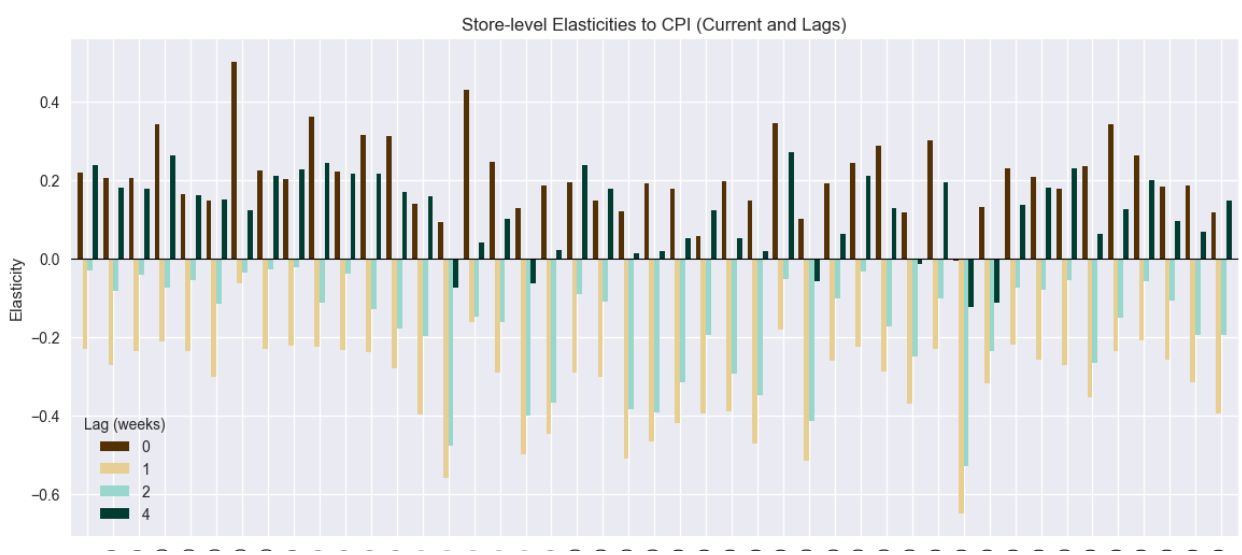

Figure 7 Lag Analysis of CPI

showing the percentage change in sales following a 1% change in CPI across multiple weeks. The first bar in each grouping represents the impact in the first week following the change, with subsequent bars representing later weeks. Comparable lag analyses for the remaining factors are provided in the Appendix, along with visualizations of the cumulative four-week impact.

Overall, both contemporaneous and lagged analyses indicate that CPI, unemployment rate, fuel price, and temperature have minimal influence on sales performance within the scope of this dataset.

Insights

The analysis suggests that inventory allocation and promotional efforts should be prioritized toward stores exhibiting the strongest growth characteristics. Among these, particular emphasis should be placed on high-growth stores that also generate high sales volumes, particularly the stores numbered 04, 13, 01, 39, and 23. These locations demonstrate strong demand as well as upward momentum, indicating a high likelihood of effectively absorbing increased inventory. For high-growth stores with comparatively lower sales volumes, including the stores numbered 41, 32, 17, 07, and 38, targeted promotional efforts may be more effective thatn inventory expansion alone. Increasing promotional activity in these locations may help accelerate the conversion of growth trends into higher overall sales performance. For stores exhibiting low growth and/or low sales, further investigation is recommended to determine the underlying causes of underperformance. This may involve evaluating external factors, such as changes in population or local economic conditions, as well as internal factors, including management effectiveness and customer experience.

Analysis of sales volatility indicates that a subset of stores exhibits unusually high variability in performance. Specifically, the stores numbered 21, 34, 28, and 38 display volatility levels exceeding 1.5 standard deviations above the mean. This level of inconsistency may be indicative of operational inefficiencies or instability at the store level. Additional investigation into these locations may help identify and address the underlying issues. In the case of store number 38, such investigation should be prioritized before implementing increased promotional efforts, given its presence in both the high growth and high volatility categories.

Seasonal analysis demonstrates that holiday periods have a substantial impact on sales performance, with most stores experiencing increases exceeding 10%, and many exceeding 20% or even 30%. Combined with the results of the economic factor analysis, this suggests that the observed increases in sales are driven primarily by holiday-related consumer behavior rather than any external economic conditions that may coincide with the holiday periods.

Finally, the analysis indicates that CPI, unemployment rate, fuel price, and temperature have negligible influence on sales performance within the scope of the dataset. As a result, these variables are unlikely to serve as meaningful predictors of store-level sales outcomes. It is more likely that sales performance is influenced by factors not captured in this analysis, such as demographic characteristics including population size and average income. Further analysis incorporating these variables is recommended to better understand the drivers of store performance.

Appendix

Growth Rates by Store

Table 3 Growth Rates by Store

Store	Growth Rate	Store	Growth Rate
1	807.83	24	-44.24
2	-404.32	25	20.12
3	344.61	26	340.18
4	2305.07	27	-1222.67
5	276.36	28	-609.76
6	-134.53	29	-111.75
7	843.52	30	-209.50
8	331.32	31	442.87
9	497.51	32	703.08
10	-896.72	33	-72.98
11	208.53	34	338.52
12	175.51	35	-2271.20
13	1182.74	36	-1381.60
14	-3275.05	37	127.30
15	-455.75	38	830.18
16	327.90	39	2371.14
17	1076.26	40	333.92
18	-755.00	41	1915.71
19	-663.67	42	396.81
20	477.85	43	-353.60
21	-442.07	44	470.75
22	-73.00	45	-49.28
23	556.25		

Models Considered

Table 4 Models Considered and Their Error Performances

Model	R^2 Score	RMSE
Linear Regression	0.166	518,343.09
Random Forest	0.966	105,120.71
Gradient Boosting	0.907	173,066.09

Lag Analysis

Week-by-Week

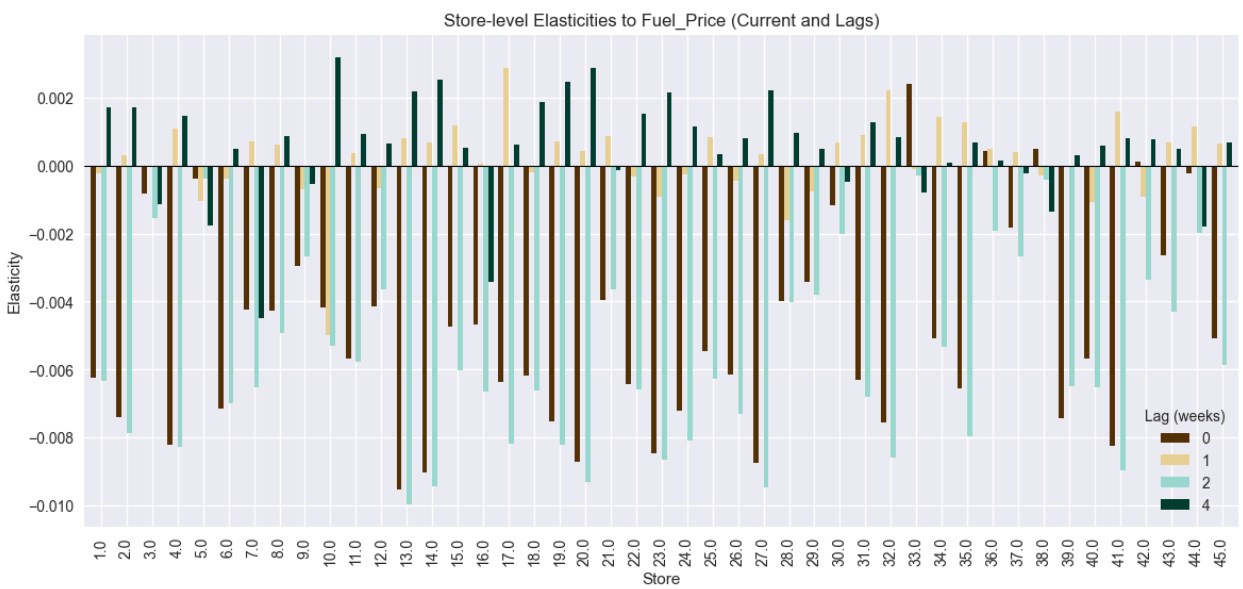

Figure 9 Lag Analysis of Fuel Price

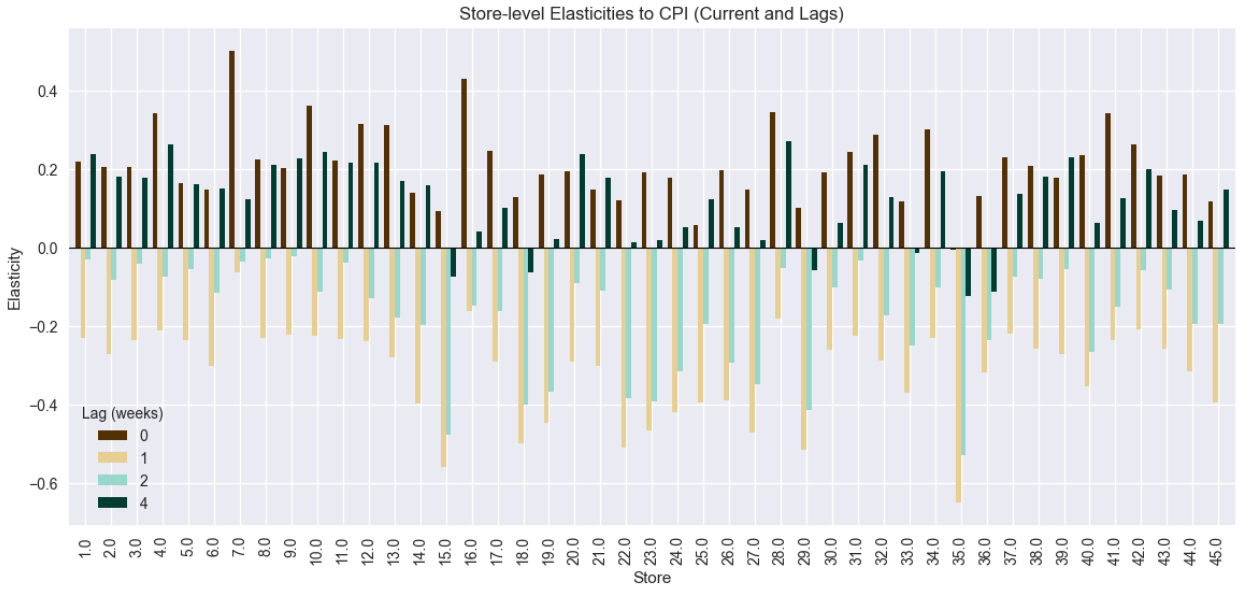

Figure 10 Lag Analysis of CPI (Repeated from Main Paper)

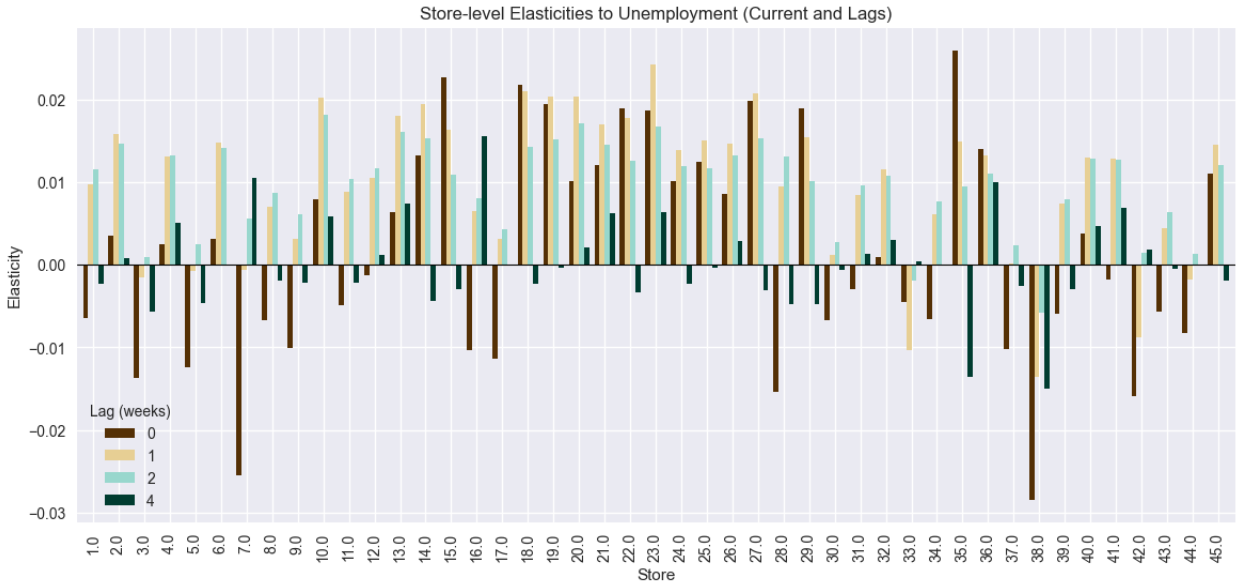

Figure 11 Lag Analysis of Unemployment Rate

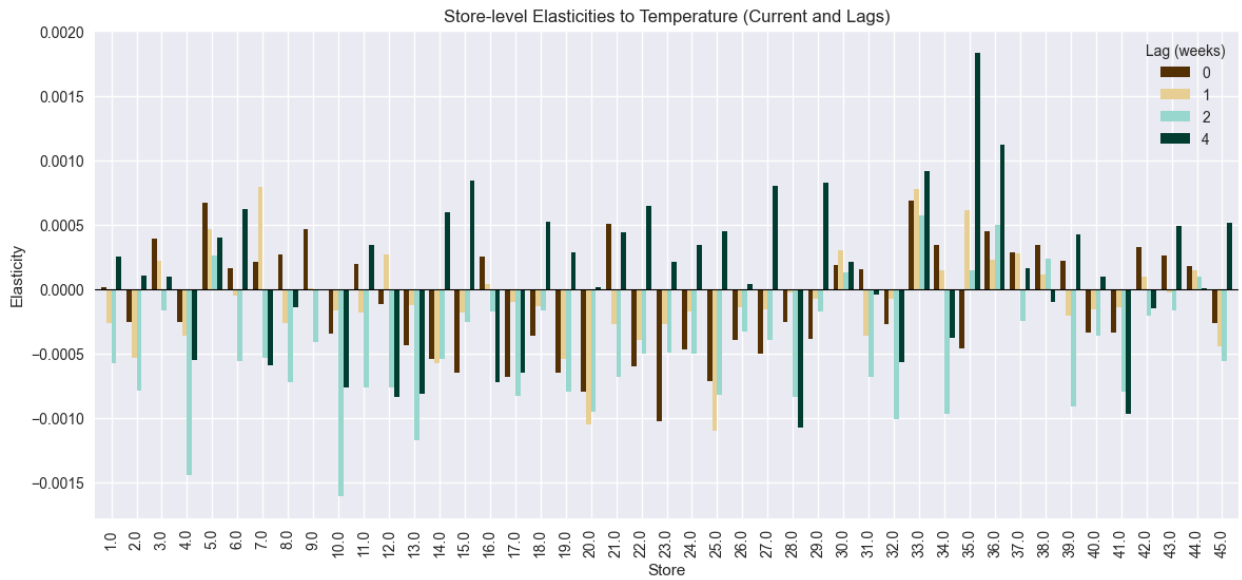

Figure 13 Lag Analysis of Temperature

Net Four Week Change

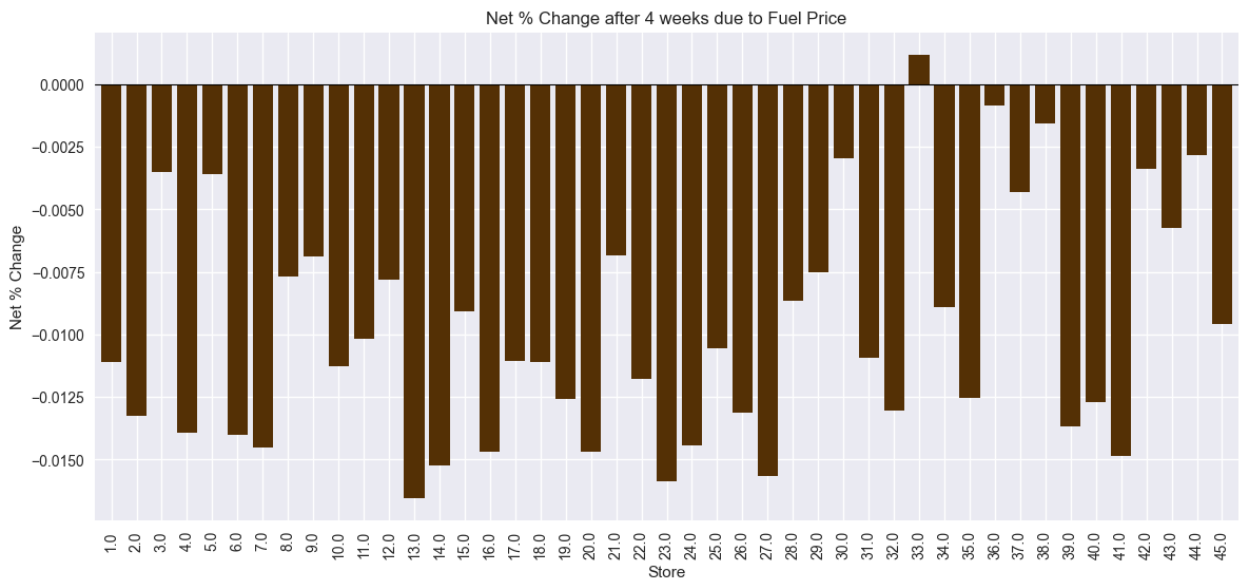

Figure 12 Net Four Week Impact of Fuel Price

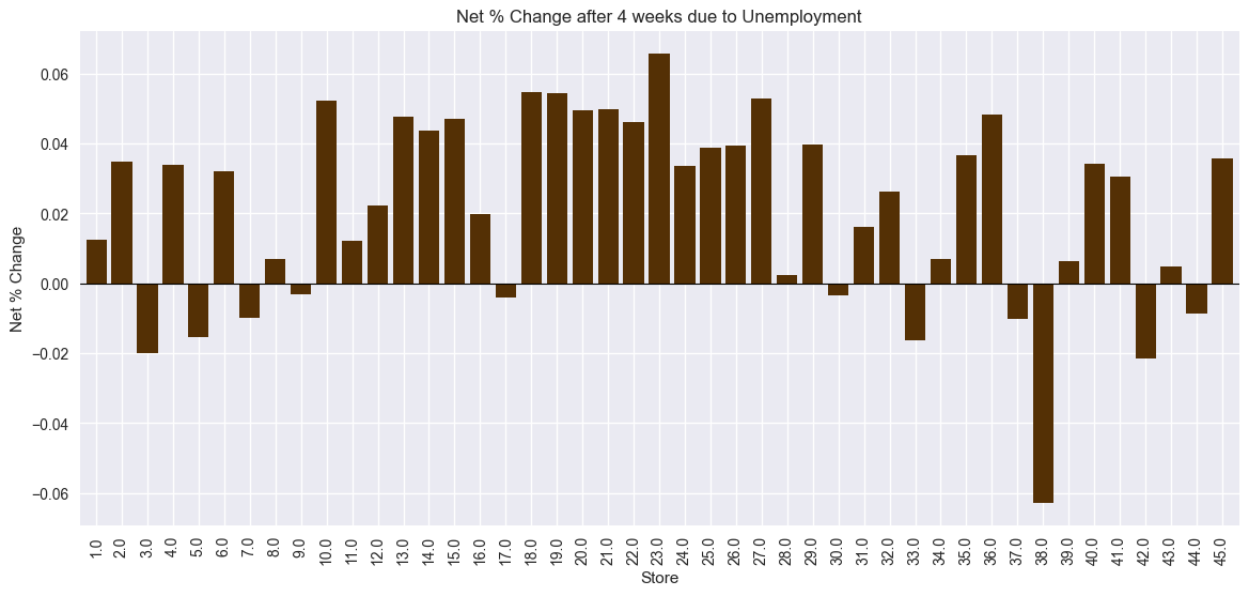

Figure 14 Net Four Week Impact of Unemployment Rate

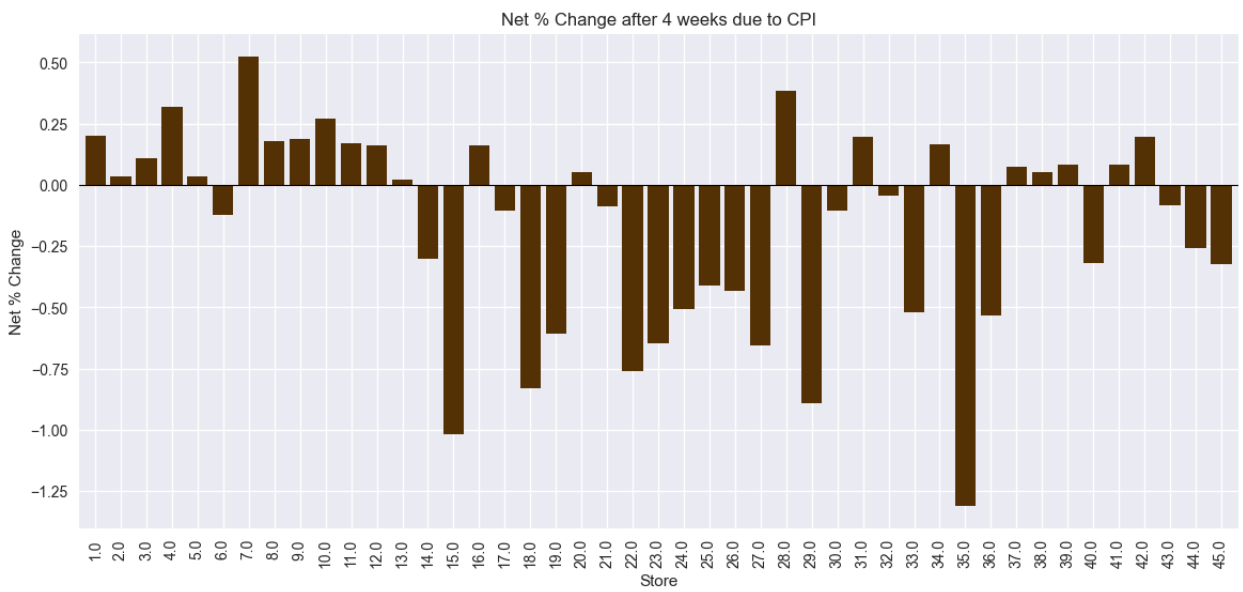

Figure 15 Net Four Week Impact of CPI

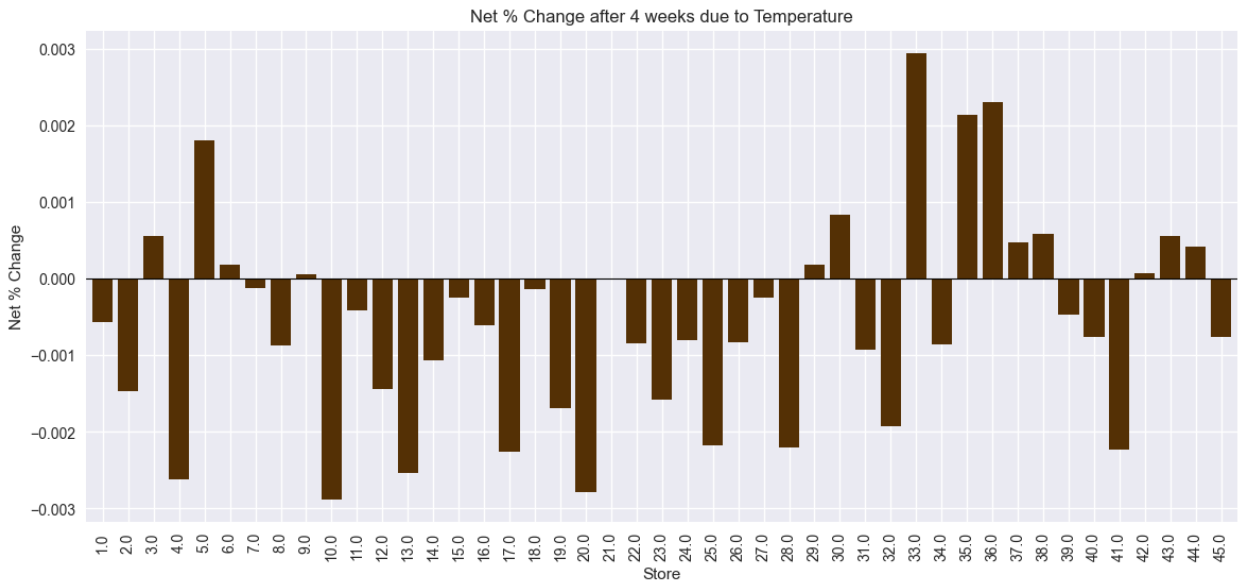

Figure 16 Net Four Week Impact of Temperature